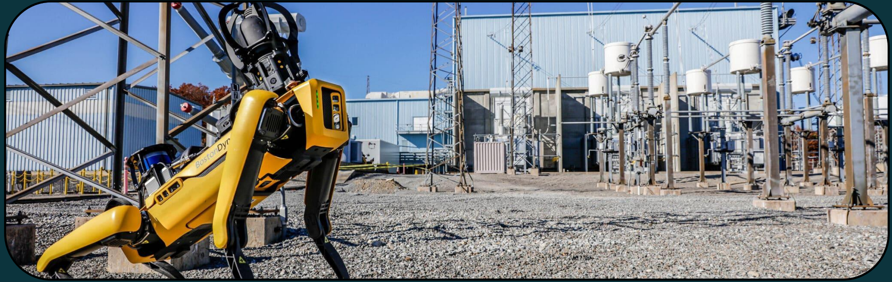
INTEGRATION IS THE HARDEST PART

Making robots applicable on-site



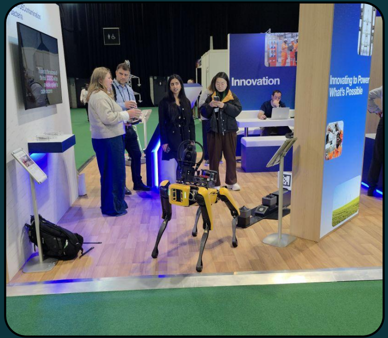
By: Daniela Shchegelskaya

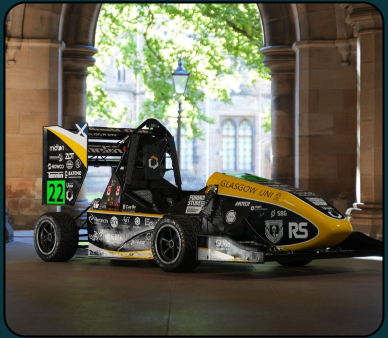
To follow along the presentation

INTRO

About Me

- Lived in 4 countries
- 2 years working at Elimpus Ltd.
- A year of working on the Spot Project
- Worked on a BrailleBot start up project and presented it to stakeholders for an Enactus competition.
- Was part of the Formula Student team at UofG, working on a racecar.
- Collection of 25 board games





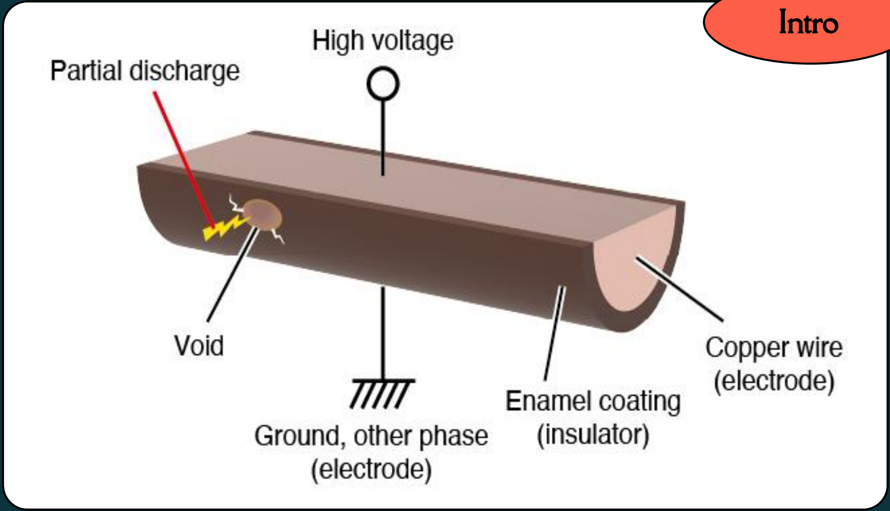
Background

What is Partial Discharge(PD) ?

Why is it so important?

What are hazardous zones at power stations?

So humans can ’t go… then what?



SPOT THE ROBOT DOG

Spot is an “off-the-shelf” yellow and black robot developed by Boston Dynamics.

It has robust capabilities but most importantly it can walk autonomously.

Can be controlled using a tablet on-site or off-site using a Boston Dynamics cloud service called Orbit.

SPOT ACCESSORIES

Tablet

Used to control Spot’s walk and camera, also for recording autowalks

Core I/O

Provides extra processing power, enhances object avoidance capabilities

Rajant Cardinal

Creates a mesh network which increases network strength

PTZ + IR Camera

360 degree camera, strong illumination and an infrared camera

Why Spot, why now?

1. Hazard Zones
2. Powering off a substation
3. Average time to solve a power station issue
4. Cost of delays
5. How much Spot costs

Problems

Everything is Perfect

Spot is so great and ready, super applicable for any site testing- why am I telling you about this?

<u>WRONG</u>

- Spot does not have GPS
- Changing networks requires manual input
- Constant updates from Boston Dynamics
- Python instead of C
- No research nor any findings to fall back on
- Spot is lazy

Network

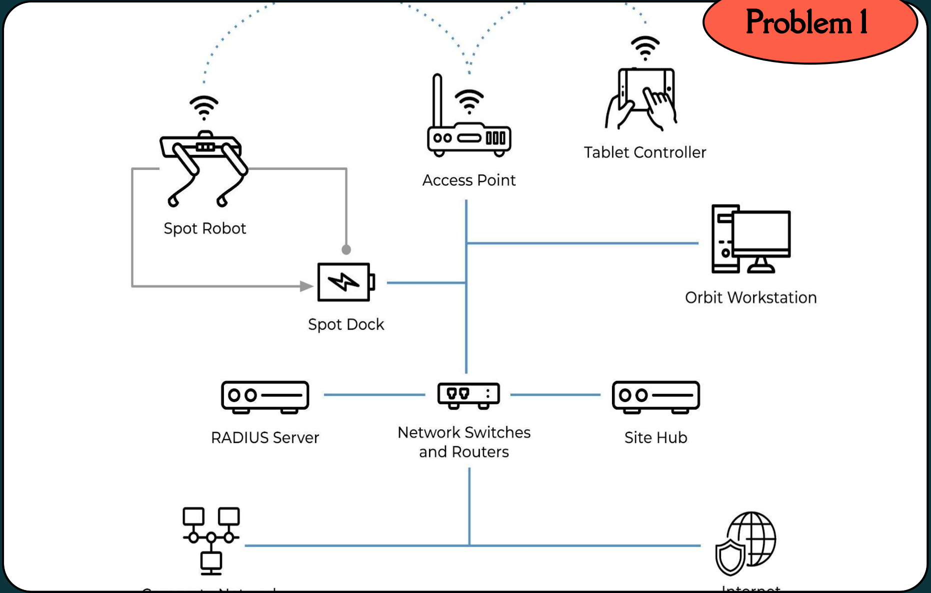
Problem 1

- Rajant Cardinal - expands network mesh to 300m
- Multiple access mode - Cloud Server and Tablet
- What is the best way to connect to it?
- No constant IP on Substations will cause issues
- Water resistant

Network

- What's the best Network Topology?
- We all think differently
- Zero Experience
- Rajant is a new product as well. Despite having Rajant Cardinal for a year there has been a massive update to hardware so we are using something quite out of date.

Problem 1



Problem 2

Programming


```
"action_id" : {...} 3 items
"data" : {} 0 items
"metadata" : { 7 items
"bosdyn.internal.point_sensor_feedback_resu
lt" :
{
3 items
"header" : { 4 items
"error" : { 1 item
```

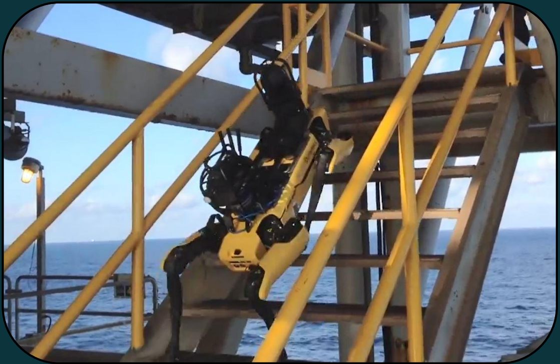
```
"clientNames" : [ 6 items
0 : "root"
1 : "bosdyn.android.spotapp"
2 : "bosdyn.android.spotapp"
3 : "mission-tick"
4 : "mission_executor"
5 : "sensor-pointing-service"
]
"epoch" : "QawUpCAIhGYHSEvB"
```


- Spot is programmed in Python
- Requires Linux
- Translating from C to Python and vice versa
- Integrating with Elimpus website
- Spot's code is in development and so is our board

Problem 3

Integration

- I am the first in the company to work on such a project
- BP and National Grid US use Spot but not the same way as we do so we are the only ones with this experience.
- This project requires a variety of experience and knowledge about various topics
- The work requires a unique approach



Problem 3

Human factors in integration

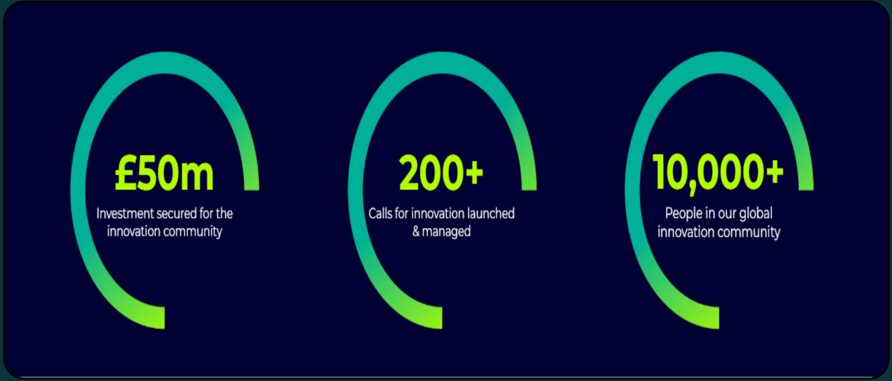
- Generation differences
- Improvement looks different for everyone
- Risk vs Experiment
- Innovation vs Stability - “if it’s not broken why fix it?
- Innovation in the energy sector
- Decision-making slows when alignment is missing.

Integration is not just an engineering task, but a negotiation of perspectives.

Improvements

Knowledge is Power

Collaboration will drive Innovation.



- More detail and development tools for products
- Improved Integration Tools

- Improved communication between companies.
- Failures are shared openly - with more transparency.

Thank you

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Q&A